

Developing a Multimodal Clinical Nursing Simulation with a Virtual Preceptor in AR

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Figure 1: Virtual preceptor feedback with multimodal sign/symptom interactions in AR clinical nursing educational simulation.

ABSTRACT

Nursing education plays a crucial role in preparing future nurses to provide high-quality patient care. To ensure its effectiveness, it is imperative to instruct nursing students in discerning patient signs/symptoms using a range of sensory modalities. Moreover, pairing nursing students with expert preceptors offers a highly effective means of providing valuable feedback and mentorship. In this paper, we propose a novel Augmented Reality (AR) simulation framework for effective clinical nursing education with multimodal signs/symptoms and a preceptor agent to provide guidance/instruction. Using the proposed framework, we developed an AR nurse training system that resembles practical clinical nursing environments.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies; Applied computing—Education—Interactive learning environments;

1 INTRODUCTION

There is a growing focus on improving the clinical competence of nursing students and professional nurses in real healthcare settings, supported by educational initiatives from nursing departments in

universities and hospitals [4, 5, 7]. This emphasis is supported by a range of educational initiatives offered by nursing departments in universities and hospitals. Physical signs and symptoms presented by patients are valuable tools for nursing education, engaging visual perception, and promoting comprehensive patient assessment [2, 3]. In addition, novice nurses undergo a crucial phase called preceptorship during their on-the-job training, where experienced nurses provide one-on-one guidance and mentorship, enhancing practical knowledge, refining clinical skills, and fostering a deeper understanding of patient care [1, 6]. AR/VR simulations provide valuable learning experiences, but the feedback given to learners is often limited to written reports or virtual agents, which may not fully replicate practical feedback in real clinical settings. However, simulating these signs and implementing preceptorship in controlled educational environments can be challenging and resource-intensive.

To address these challenges, we propose a novel AR clinical nursing educational simulation system AR/VR-based nursing education and literature (see Figure 1). The system we create enables learners (both nursing professionals and students) to engage in simulations within an AR environment that faithfully replicates real hospital clinical training. Learners can immerse themselves in diverse patient signs/symptoms using intuitive interactions, such as hand gestures and speech. Within the AR simulation environment, learners can interact with a virtual patient displaying various multimodal signs and symptoms, such as body temperature, breathing, and smell. To enhance the learning experience, we incorporate an engaging virtual preceptor with auto-generated feedback capabilities. As learners navigate the nursing simulation, the system continuously generates real-time feedback based on their performance scores, delivered by the virtual preceptor in a visual-audio format. Our research sheds light on its strengths and areas for improvement, highlighting its promising role in advancing nursing education and clinical training.

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2 NURSING INTERVENTION FRAMEWORK FOR MULTIMODAL PATIENT SIGN/SYMPTOM AND VIRTUAL PRECEPTOR

2.1 Framework Overview

We propose a novel framework for training nursing professionals and students in AR clinical nursing simulation incorporating multimodal sign/symptom interactions and the virtual preceptor. The proposed framework consists of three main parts, aiming at providing scenario-based clinical nursing education and assessment in simulated nursing education settings: (1) multimodal patient sign/symptom interaction, and (2) automatic feedback generation based on a virtual preceptor.

2.2 Multimodal Patient Sign/Symptom Interaction

The proposed AR clinical simulation framework enables natural user inputs, allowing learners to interact with physical and virtual objects, e.g., a virtual patient and nursing tools, in the scene using hand gestures and speech. Furthermore, the framework notably addresses multimodal patient signs and symptoms, a distinctive feature that enhances the realism and diversity of nursing intervention experiences. For example, we represent patient conditions using a variety of multimodal signs/symptoms including body temperature, respiration, and sense of smell, in combination with real devices and the virtual patient. These multimodal signs/symptoms provide learners with various opportunities to explore actual patient conditions beyond verbal/behavioral cues. In nursing education, while experienced nurses can accurately recognize patients' signs and symptoms and provide effective nursing care, novice nurses or students who have little experience in dealing with actual patients may not be able to easily recognize and digest the same patient's signs and symptoms.

2.3 Automatic Generation of Virtual Preceptor Feedback

This framework is actively employed in the field of clinical nursing education, based on the principles of preceptorship. What sets preceptorship apart from other forms of education is that it occurs in actual clinical settings, with a 1:1 relationship between a senior nurse and a new nurse, allowing real-time feedback through observation or personalized instruction.

Generally, the preceptor can provide guidance and feedback to the preceptee (learner) from various perspectives and timings. In this study, we ensure that the learner receives feedback on what they have learned without any other interventions or disruptions. This is to enable independent learning by preventing the learner from being disturbed during nursing interventions. By undergoing self-directed nursing intervention performance and receiving detailed feedback from the preceptor, we aim to enable nurses to perceive patients' signs/symptoms with critical thinking and carry out nursing interventions accordingly. Furthermore, by using a virtual preceptor, we aim to maximize its effectiveness by providing personalized feedback tailored to the learner and simultaneously displaying a checklist (feedback list) of the tasks performed, allowing for a visual representation.

3 PEDIATRIC POSTOPERATIVE NURSING SIMULATION

Based on the proposed framework for AR clinical nursing educational simulation using multimodal patient signs/symptoms, we develop the system with the specific clinical nursing educational scenario made with nursing education faculties in our university. Pediatric clinical nursing, which is one of the critical nursing procedures where confirming multimodal patient signs/symptoms is vital, was chosen as the base. To develop the AR clinical nursing environment including the expression of multimodal symptoms, we also describe the system apparatus and how to use them. Through this, we aim to verify the learning efficacy of the system based on the proposed framework as an educational tool for the participating nursing experts and to comparatively analyze the system's evaluation accuracy.

3.1 Simulation Scenario and Implementation

The AR clinical nursing scenario that we designed is about pediatric appendicitis postoperative nursing. There are nursing interventions that the learner should perform during the simulation with simple interaction such as 'Inspection of abdominal surgical site' and 'Nursing intervention against vomiting', and multimodal signs/symptoms including body temperature, respiration, and olfactory not only visual-audio. The virtual preceptor provides feedback based on the nursing interventions conducted.

In the development, the system was developed using Unity3D. The AR environment was built using the Meta Quest 2 headset, which received input data from participants' hand gestures and speeches. Additionally, a StereoLabs ZED Mini camera was attached to the Meta Quest 2 to render real-world scenes in real-time and overlay virtual objects on physical objects by recognizing markers. Arduino was used for each piece of equipment to express multimodal signs/symptoms, and all interactions were set up to operate automatically within the system through a Bluetooth connection between Unity3D and Arduino.

4 CONCLUSION

This study proposes a method to simulate a real clinical nursing practice environment by combining virtual patients including multimodal signs/symptoms and a preceptor agent through an AR-based system. The system is implemented to fully and automatically control multimodal signs/symptoms, and the virtual preceptor agent has realized the ability to provide real-time feedback to participants. We expect our framework with multimodal signs/symptoms to increase critical thinking ability in clinical nursing, and the preceptor agent to increase the degree of immersion in learning, particularly for novice nurses. However, several areas need to be improved, such as the strength and quality of the sensor, and the functionality of the virtual preceptor. In future studies, interaction with virtual patients, realism, and immersion should be further improved, and existing constraints will be addressed to pursue the continuous development and efficiency of AR tools in nursing education.

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